

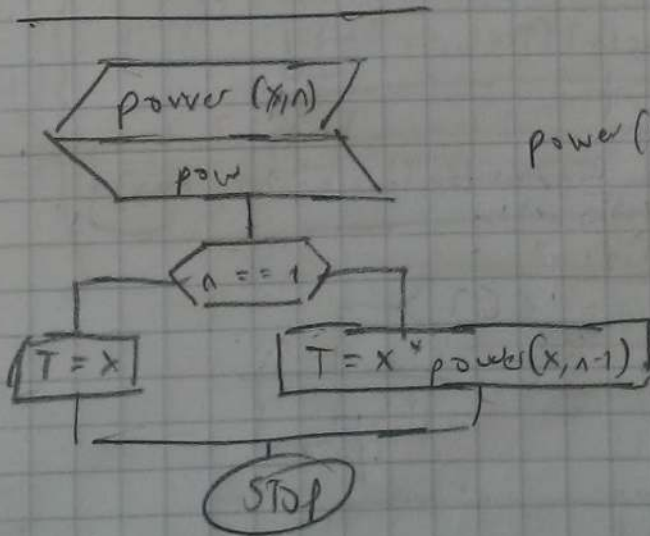
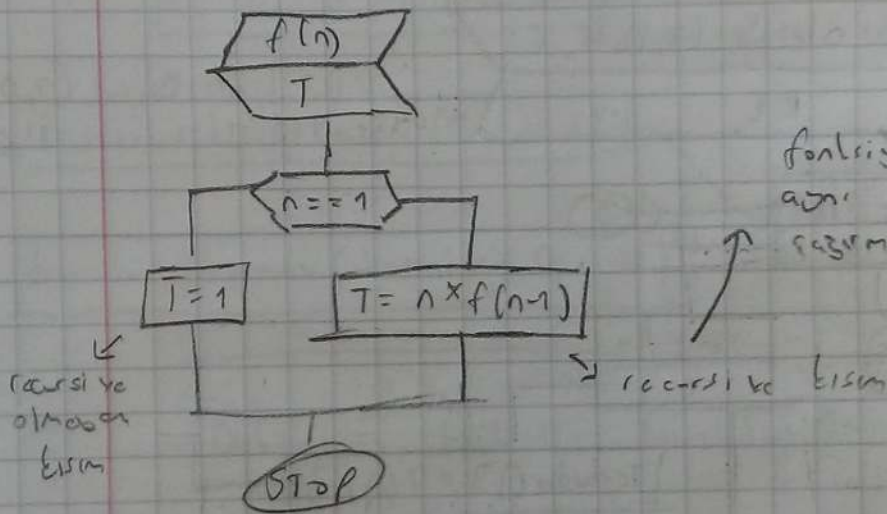
Recursive (Özyinekerli) Fonksiyonlar:

$a \rightarrow a$ (direkt)

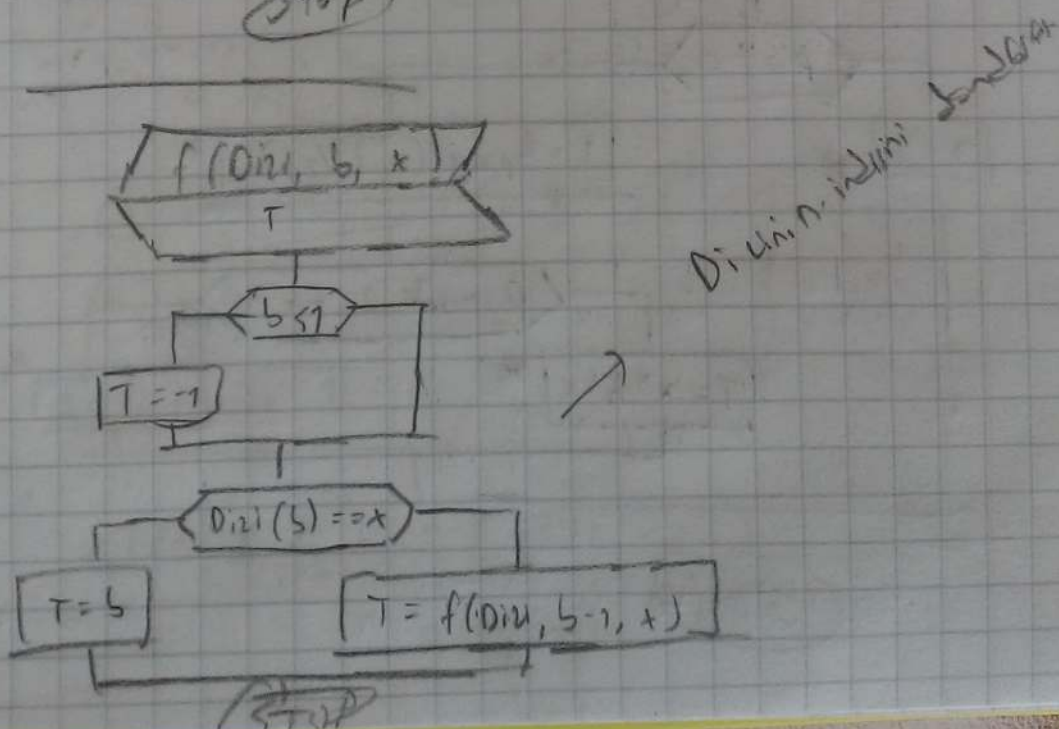
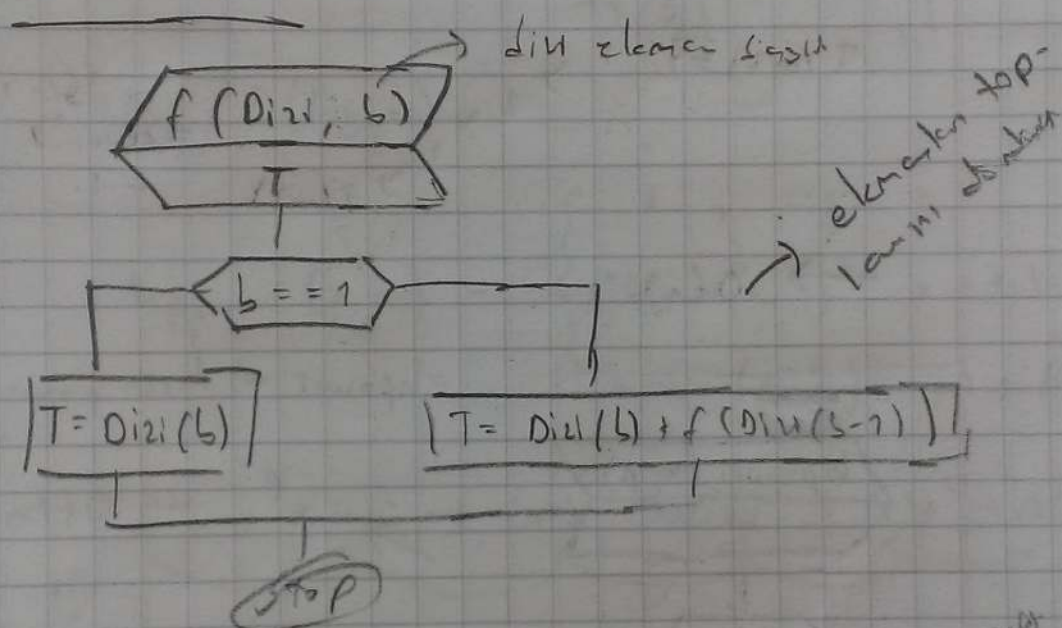
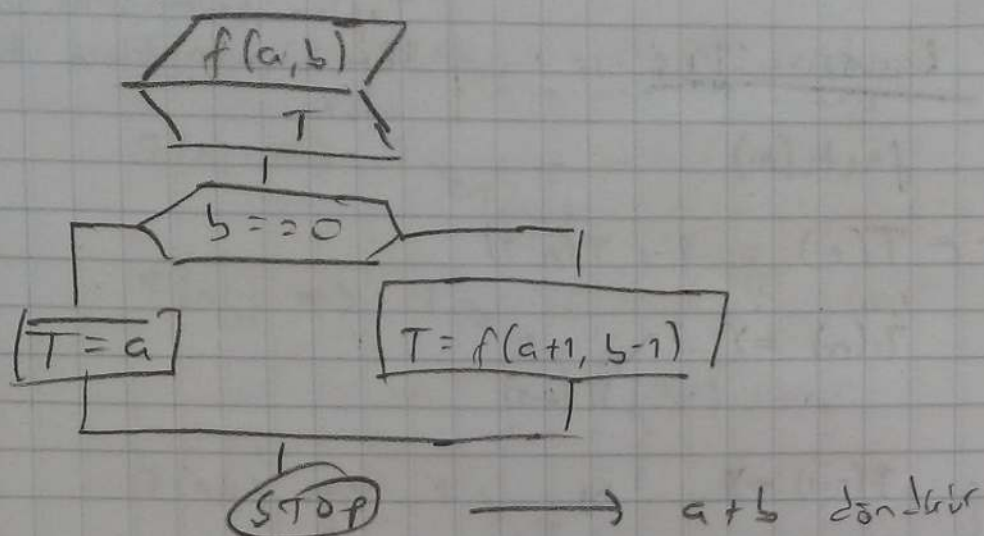
$a \rightarrow b \rightarrow a$ (dolaylı)

faktöriyel:

$$n = n \times n-1$$
$$\rightarrow f(n) = n \times f(n-1)$$



$$\text{power}(x, n) = x \times x^{n-1}$$



Recessive trait

Toplam barınaklık ücreti
tüm elementlerinin toplamı
olur.

$$\text{fact}(x) :$$

$$T(n) = 1 + T(n-1)$$

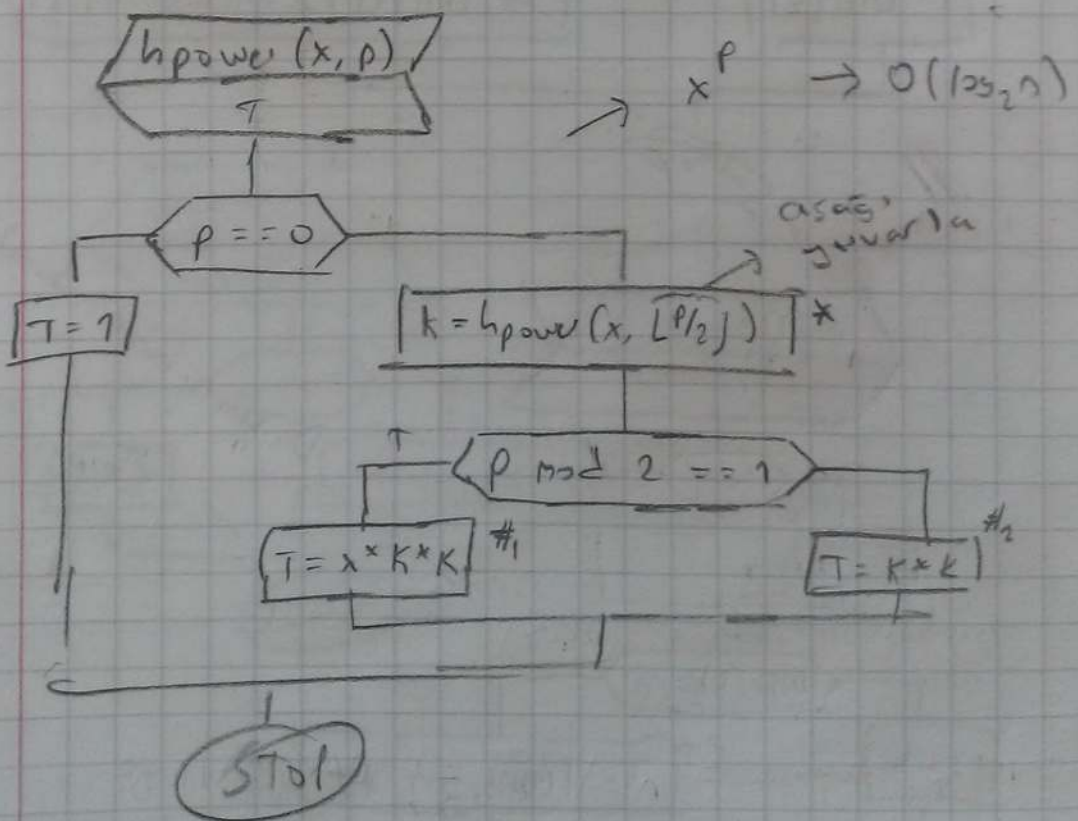
$$T(n) \Rightarrow 1 \rightarrow T(n-1)$$

$$T(n-1) \Rightarrow 1 \rightarrow 1 \rightarrow T(n-2)$$

$$T(n-2) \Rightarrow T(n-3)$$

$$\rightarrow O(n)$$

low power $\rightarrow$ high power



$$T(n) = 1 + T(n/2)$$

$$T(n) \Rightarrow 1 \rightarrow T(n/2)$$

$$T(n/2) \Rightarrow 1 \rightarrow 1 \rightarrow T(n/4)$$

$$T(n/4) \Rightarrow 1 \rightarrow 1 \rightarrow 1 \rightarrow T(n/8)$$

$$\begin{array}{c} \uparrow \\ \vdots \\ \uparrow \\ h = \log_2 n \\ \downarrow \\ \vdots \\ \downarrow \end{array}$$

$$T(n) = O(\log_2 n)$$

$$h_{\text{power}}(x, p) = \begin{cases} 1, & p = 0 \\ x^{h_{\text{power}}(x, p/2)^2}, & p \rightarrow \text{t.c.} \\ h_{\text{power}}(x, p/2)^2, & p \rightarrow \text{s.11} \end{cases}$$

$$x^{32} \rightarrow (x^{16})^2 \rightarrow (x^8)^2 \rightarrow (x^4)^2 \rightarrow (x^2)^2 \rightarrow (x^1)^2$$

$$\log_2 32 = 5 \text{ işlem}$$

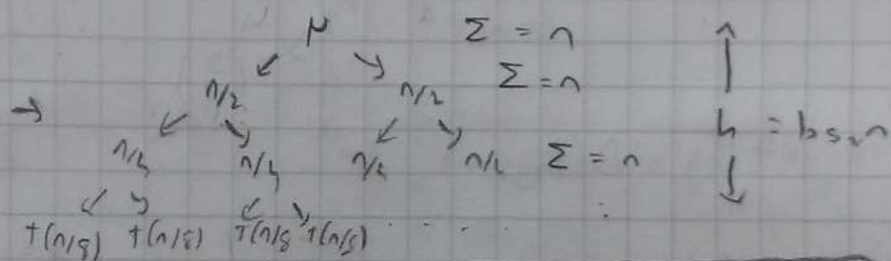
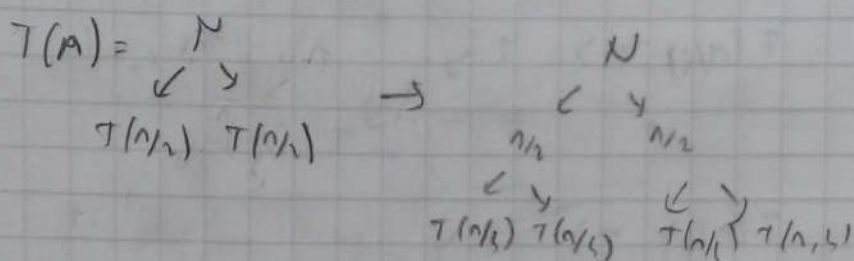
$$\left. \begin{array}{l} \textcircled{*} \rightarrow k = h_{\text{power}}(x, \lfloor p/2 \rfloor)^2 \\ \textcircled{H_1} \rightarrow T = x * k \\ \textcircled{H_2} \rightarrow T = k \end{array} \right\} \begin{array}{l} \text{olur i2i} \\ T(n) = 1 + 2T(n/2) \\ \text{olurdu} \end{array}$$

$$T(n) = \sum_{i=0}^{\log_2 n} 2^i = 2 \cdot 2^{\log_2 n - 1}$$

$$= O(n)$$

$$h = \log_2 n$$

~~2.8.81~~ $T(n) = 2T(n/2) + n$ bir ağaç olsun:
 ↓
 quick sort'un merge sort'a göre

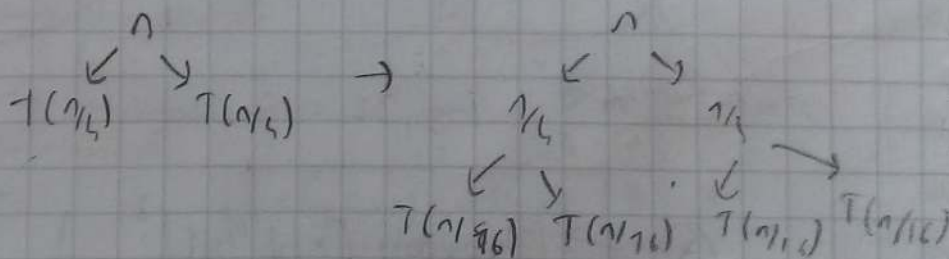


$$T(n) = n \cdot \log_2 n$$

~~2.8~~

$$T(n) = 2T(n/4) + n$$

$$\sum_{i=0}^n A^i = \begin{cases} \frac{A^{n+1} - 1}{A - 1} & (A > 1) \\ \frac{1 - A^{n+1}}{1 - A} & (A < 1) \end{cases}$$



$$h = \log_4 n$$

$$T(n) = n + \frac{n}{2} + \frac{n}{4} + \frac{n}{8} \dots$$

$$= n \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right) = n \cdot \sum_{i=0}^{\log_4 n} \left(\frac{1}{2} \right)^i$$

$$O(n)$$

$$= n \cdot \frac{1 - \left(\frac{1}{2} \right)^{\log_4 n + 1}}{1 - \frac{1}{2}} = 2n - \sqrt{n}$$

~~$$T(n) = 8T(n/2) + n$$~~

$$\begin{array}{c} n \\ \swarrow \quad \searrow \quad \vdots \\ T(n/2) \quad T(n/2) \quad \dots \end{array} \rightarrow 8 \text{ times } T(n/2)$$

$$\begin{array}{c} n \\ \swarrow \quad \searrow \quad \vdots \\ n/2 \quad n/2 \quad \dots \end{array} \rightarrow 8 \text{ times } n/2$$

$$\begin{array}{c} \swarrow \quad \searrow \quad \vdots \\ T(n/4) \quad T(n/4) \quad \dots \end{array} \rightarrow 64 \text{ times } T(n/4)$$

$$\rightarrow 64 \text{ times } \frac{n}{4}$$

$$h = \log_2 n$$

$$T(n) = n + 8 \cdot \frac{n}{2} + 64 \cdot \frac{n}{4} + \dots$$

$$= n + 4n + 8n + 16n$$

$$= n (1 + 4 + 8 + 16 + \dots)$$

$$T(n) = n \cdot \sum_{i=0}^{\log_2 n} (4^i)$$

$$= n \cdot \frac{4^{\log_2 n + 1} - 1}{4 - 1} = n \cdot \frac{4 \cdot n^{\log_2 4} - 1}{3}$$

$$= \frac{4}{3} n^3 - \frac{n}{3}$$

$$\rightarrow O(n^3)$$

Ex

$$T(n) = 8T\left(\frac{n}{2}\right) + \frac{n^2}{1}$$

$$T\left(\frac{n}{2}\right) \xleftarrow{\frac{n^2}{1}} \dots 8 \text{ times } T\left(\frac{n}{2}\right)$$

$$\begin{array}{l} \xleftarrow{\frac{n^2}{1}} \left(\frac{n}{2}\right)^2 \xleftarrow{\frac{n^2}{1}} \left(\frac{n}{2}\right)^2 \dots \rightarrow 8 \text{ times } \left(\frac{n}{2}\right)^2 \\ \xleftarrow{\frac{n^2}{1}} T\left(\left(\frac{n}{4}\right)^2\right) T\left(\left(\frac{n}{4}\right)^2\right) \dots \rightarrow 16 \text{ times } \left(\frac{n}{4}\right)^2 \\ \dots \rightarrow 64 \text{ times } \left(\frac{n}{8}\right)^2 \end{array}$$

$$h = \log_2 n$$

$$T(n) = n^2 + 2n^2 + 4n^2 \dots = n^2(1+2+4 \dots)$$

$$= n^2 \cdot \sum_{i=0}^{\log_2 n} 2^i = n^2 \cdot \frac{2^{\log_2 n + 1} - 1}{2 - 1} = \frac{2 \cdot 2^{\log_2 n} - 1}{1} n^2$$

$$= (n-1) \cdot n^2 \rightarrow O(n^3)$$

EX

$$T(n) = \underbrace{8}_{\text{recursive kism}} T(n/2) + \underbrace{1}_{\text{recursive almanchen}}$$

$$\left(\frac{n}{2}\right) \rightarrow 1$$

$$\downarrow$$

$$\left(\frac{n}{2}\right)^0$$

$$\begin{array}{c} 1 \\ \swarrow \downarrow \\ T(n/2) \quad T(n/2) \dots \end{array} \rightarrow 8 \text{ times } T(n/2)$$

$$\begin{array}{c} 1 \\ \swarrow \downarrow \\ 1 \quad 1 \dots \end{array} \rightarrow 8 \text{ times } 1$$

$$\begin{array}{c} 1 \\ \swarrow \downarrow \\ 1 \quad 1 \dots \end{array} \rightarrow 8 \text{ times } \left(\frac{n}{2}\right)^0$$

$$\begin{array}{c} \swarrow \downarrow \\ T(n/2) \quad T(n/2) \dots \end{array} \rightarrow 8 \text{ times } T(n/2)$$

$$\begin{array}{c} 1 \\ \swarrow \downarrow \\ 1 \quad 1 \dots \end{array} \rightarrow 8 \text{ times } 1$$

$$h = \log_2 n$$

$$T(n) = 1 + 8 + 64 + \dots$$

$$= \sum_{i=0}^{\log_2 n} (8)^i = \frac{8^{\log_2 n + 1} - 1}{8 - 1}$$

$$= \frac{8 \cdot n^{\log_2 8} - 1}{7} = \frac{8n^3 - 1}{7} \rightarrow O(n^3)$$

Ex

$$T(n) = 4T(n/3) + 1$$

$$\begin{array}{c} 1 \\ \swarrow \searrow \\ T(n/3) \quad T(n/3) \dots \end{array} \rightarrow 4 \text{ times } T(n/3)$$

$$\begin{array}{c} 1 \\ \swarrow \searrow \\ 1 \quad 1 \dots \end{array} \rightarrow 4 \text{ times } 1$$
$$\rightarrow 16 \text{ times } 1$$

$$h = \log_3 n$$

$$T(n) = \sum_{i=0}^{\log_3 n} 4^i = \frac{4^{\log_3 n + 1} - 1}{4 - 1}$$

$$\leftarrow = \frac{4 \cdot n^{\log_3 4} - 1}{3}$$
$$O(n^{\log_3 4})$$

EX

$$T(n) = 9 T(n/3) + 1$$

$$\begin{array}{c} 1 \\ \swarrow \searrow \\ T(n/3) \dots \end{array} \rightarrow 9 \text{ times } T(n/3)$$

$$\begin{array}{c} 1 \\ \swarrow \searrow \\ 1 \quad 1 \dots \end{array} \rightarrow 9 \text{ times } 1$$
$$\rightarrow 81 \text{ times } 1$$

$$L = \log_3 n$$

$$T(n) = \sum_{i=0}^{\log_3 n} 9^i = \frac{9^{\log_3 n + 1} - 1}{9 - 1} = \frac{9 \cdot n^{\log_3 9} - 1}{8}$$
$$= \frac{9n^2 - 1}{8} \rightarrow O(n^2)$$

EX

$$T(n) = 4 T(n/2) + n$$

$$\begin{array}{c} \wedge \\ \swarrow \searrow \\ T(n/2) \quad T(n/2) \dots \end{array} \rightarrow 4 \times T(n/2)$$

$$\begin{array}{c} \wedge \\ \swarrow \searrow \\ n/2 \quad n/2 \dots \\ \swarrow \searrow \\ T(n/4) \quad T(n/4) \dots \end{array} \rightarrow 16 \times T(n/4)$$

$$h = \log_3 n$$

$$T(n) = n + 4 \cdot \frac{n}{3} + 16 \cdot \frac{n}{9} \dots$$

$$= n \left(1 + \frac{4}{3} + \frac{16}{9} \dots \right)$$

$$= n \cdot \sum_{i=0}^{\log_3 n} \left(\frac{4}{3}\right)^i = n \cdot \frac{\left(\frac{4}{3}\right)^{\log_3 n + 1} - 1}{\frac{4}{3} - 1}$$

$$= n \cdot \frac{\frac{4}{3} \cdot \frac{4^{\log_3 n}}{3^{\log_3 n}} - 1}{\frac{1}{3}}$$

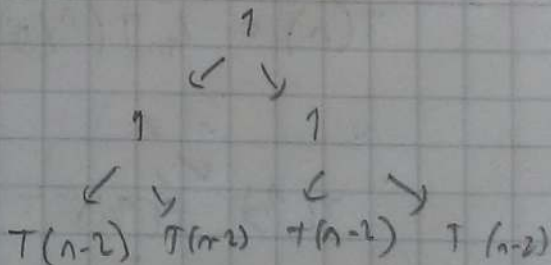
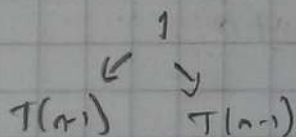
$$= 3n \cdot \left(\frac{4 \cdot 4^{\log_3 n}}{3^n} - 1 \right)$$

$$= 4 \cdot 4^{\log_3 n} - 3n \rightarrow O(n^{\log_3 4})$$

$$2 \cdot T(n-2) + 1$$

EX

$$T(n) = 2T(n-1) + 1$$



$$h = n$$

$$T(n) = 1 + 2 + 4 + \dots$$

$$\approx \sum_{i=0}^n 2^i = \frac{2^{n+1} - 1}{2 - 1} = 2 \cdot 2^n - 1 \rightarrow O(2^n)$$

n^2	2^n	
1	1	2
16	100	1024
20	400	1M
30	900	1B ...

} $2^n - n^2$
Kasıtlıdır mesela

$$T(n) = T(n-1) + T(n-2) + 1 \rightarrow \text{Fibonacci}$$

$$\begin{array}{c} 1 \\ \swarrow \quad \searrow \\ T(n-1) \quad T(n-2) \end{array}$$

$$\begin{array}{c} 1 \\ \swarrow \quad \searrow \\ 1 \quad 1 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ T(n-2) \quad T(n-3) \quad T(n-3) \quad T(n-4) \dots \end{array}$$

→ Ağacın kökleri farklı zamanlarda olduğundan

$$T(n) = 2 + (n-1) + 1 \text{ v.s.} \rightarrow O(n^2)$$

5 binden daha büyük olmalı

$$\text{Gerekli} \rightarrow O(k^n), \quad k = \frac{1+\sqrt{5}}{2} \quad (\text{altın oran})$$

ALTERNATIVE
sol

$$F = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

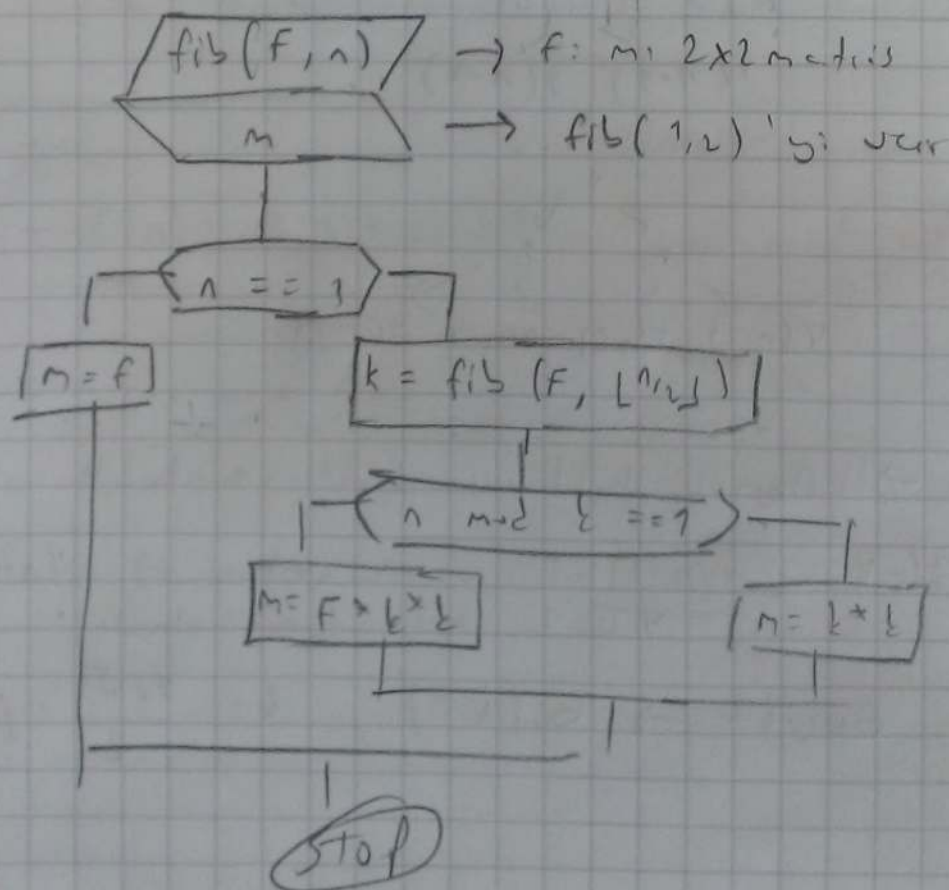
$$F^2 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$F^3 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$$

$$F^4 = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$$

$$F^5 = \begin{bmatrix} 8 & 5 \\ 5 & 3 \end{bmatrix}$$

$$F^n = \begin{bmatrix} \text{fib}(n+1) & \text{fib}(n) \\ \text{fib}(n) & \text{fib}(n-1) \end{bmatrix} \rightarrow \text{by power series } (O(\log_2 n))$$



$$T(n) = T(n/2) + C \rightarrow \text{Önemsiz işlen sayısı, (n'den bağımsız)}$$

$$= O(\log_2 n)$$

Sepet Algoritması :

Sipariş	Sepet ID	Ürün ID
1 →	2	42
2 →	3	15
⋮	3	42
⋮	⋮	⋮
N →	5	6
	5	15

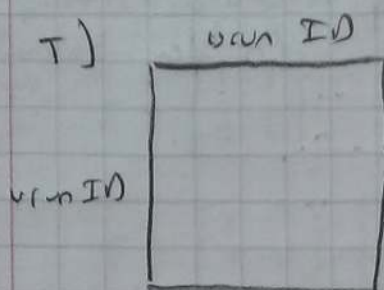
A)

	Sepet ID	↑ $sep(r)$
	0 0 1 1 0 0 1 0 0	
	1	
	0	
	1	
	0	
	0	
	1	
	0	
Ürün ID		
↓ $ür(r)$		

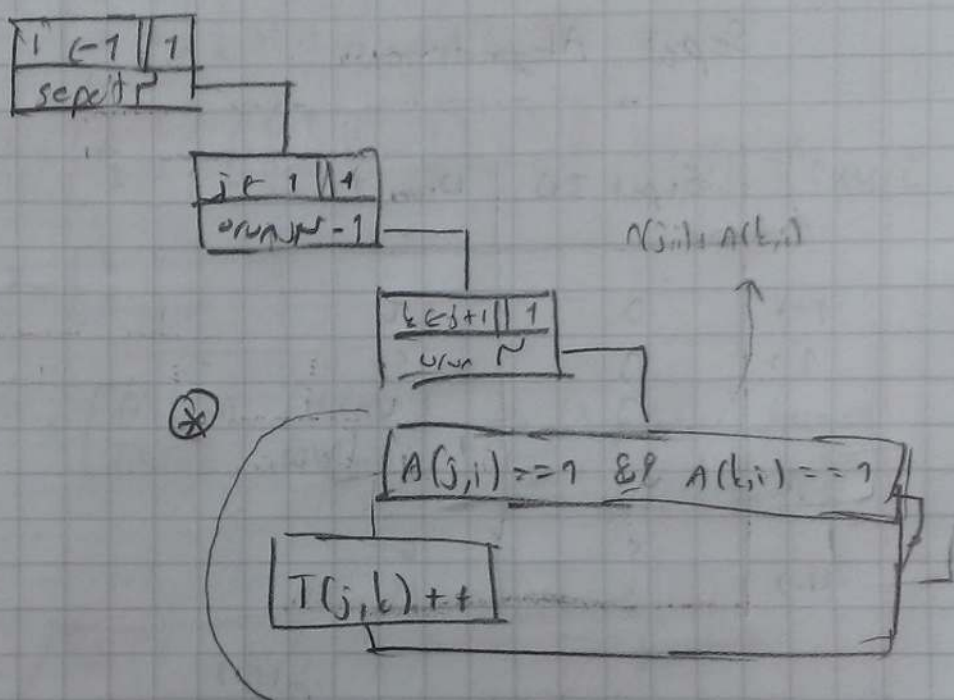
$i \leq 1$	1
r	

$$A(s(i,2), s(i,1)) = 1$$

S 'den A'ya geçiş?



→ A'dan T'ye geçile!



⊗ → $T(j,k) += A(j,i) * A(k,i)$
(çigileştirme)

Buna da ibisi Apriori algoritması

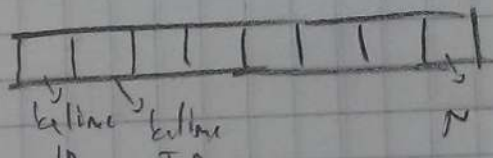
Sıfırlama Alg.

	1	2	3	4	5
1	'1'	'2'	'3'	'4'	'5'
2	-	-	-	-	-
3	'1'	'1'	-	-	-
4	-	-	-	-	-
5	-	-	-	-	-

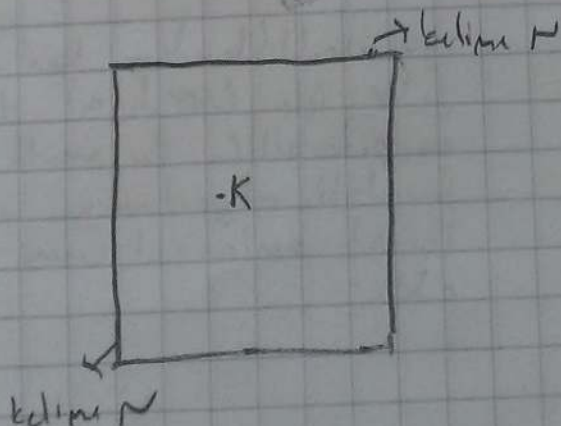
$A B = CD$
 $\downarrow \downarrow \downarrow \downarrow \downarrow$
 12 13 32 14 15 ...

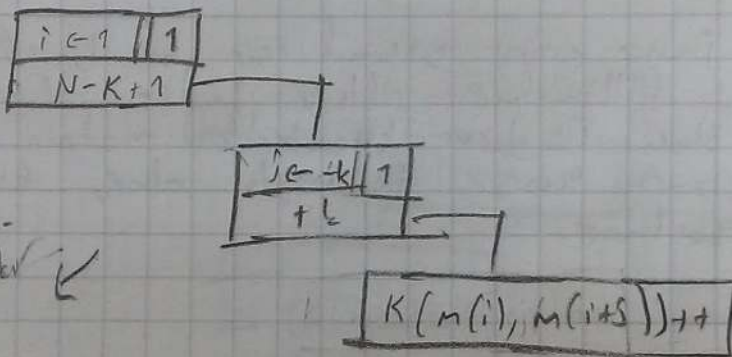
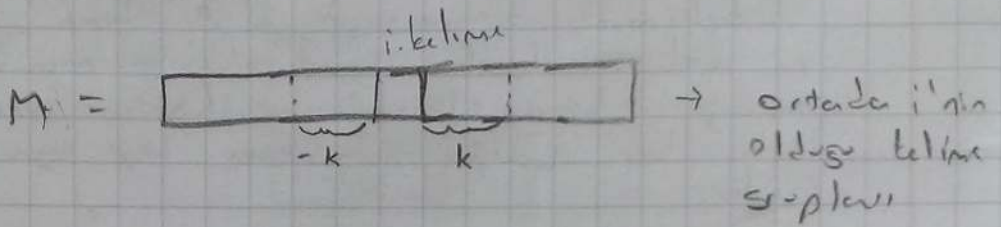
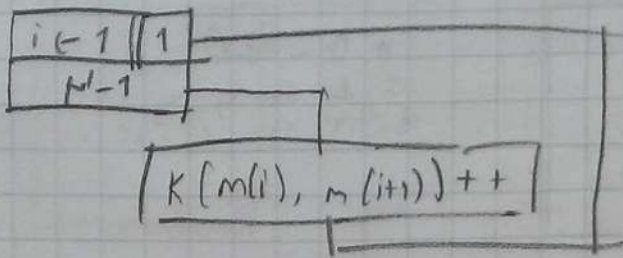
Sıfırlamanın bir amacı da, "A" harfi girildiğinde çok kullanılan harf olması mantığıyla mesaj tahmini olarak yapılabilir. Bunun sebebi bir harfin numarasına hep aynı olanıdır.

Bunun yerine sadece işin bir harfin bir dilde kullanılması sıklığı kadar farklı numarası olan B5 gibi kısıtlı olacak metin tüm sayıların birer frekansı olur, kıtlığı zorlanır.

$M =$  $\rightarrow$ metin
 (bu kelime için 10 var)

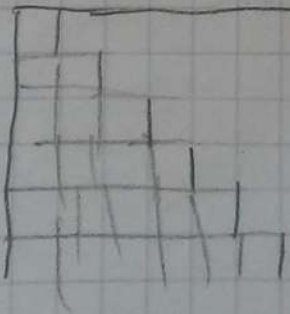
Arka arkaya en sık tekrar eden kelime ikilisi?





kelime grup-
larının içindeki
aralamsal
gereklilik

Harita → sabitli arası zamanlı



bu veritabanından her bir
şehir için koordinat tablosu
oluşturulabilir. Bu mantık ya-
rımadaki algoritmaya göre
her şehir arasındaki uzaklık
elde edilir.